

THE COMMONS - Compact Edition

VRChat Social / Bar / Cafe / Academic / DJ Venue
Design & Implementation Packet v0.1



Reference image = mood/composition. CAD/spec = authoritative geometry.

1. Project brief

THE COMMONS is a compact, vertically layered metaverse “third place.” Its purpose is to make conversation, academic exchange, casual drinking/cafe use, poster demos and DJ nights feel like different intensities of the same social ecosystem rather than separate rooms stitched together.

- Footprint: 28.0 m x 18.0 m (504 m² gross 1F envelope).
- Levels: 1F +0.000 m, 2F +4.800 m, roof +9.600 m.
- Comfortable population: 32; event target: 48; design max: 64.
- Central double-height atrium is the primary orientation and social volume.
- Visual language: dark neo-industrial structure + warm hospitality lighting + restrained cyan/blue media light + flat-light shading.

2. Concept / world setting

The venue is imagined as a former urban technical exchange and observatory that has been informally converted into a public salon. The old media core survives as a central screen/hologram system. Researchers, makers, artists and DJs use the same room at different times. The world should feel like a place people return to, not an event set that only works when a host is present.

Experience sequence

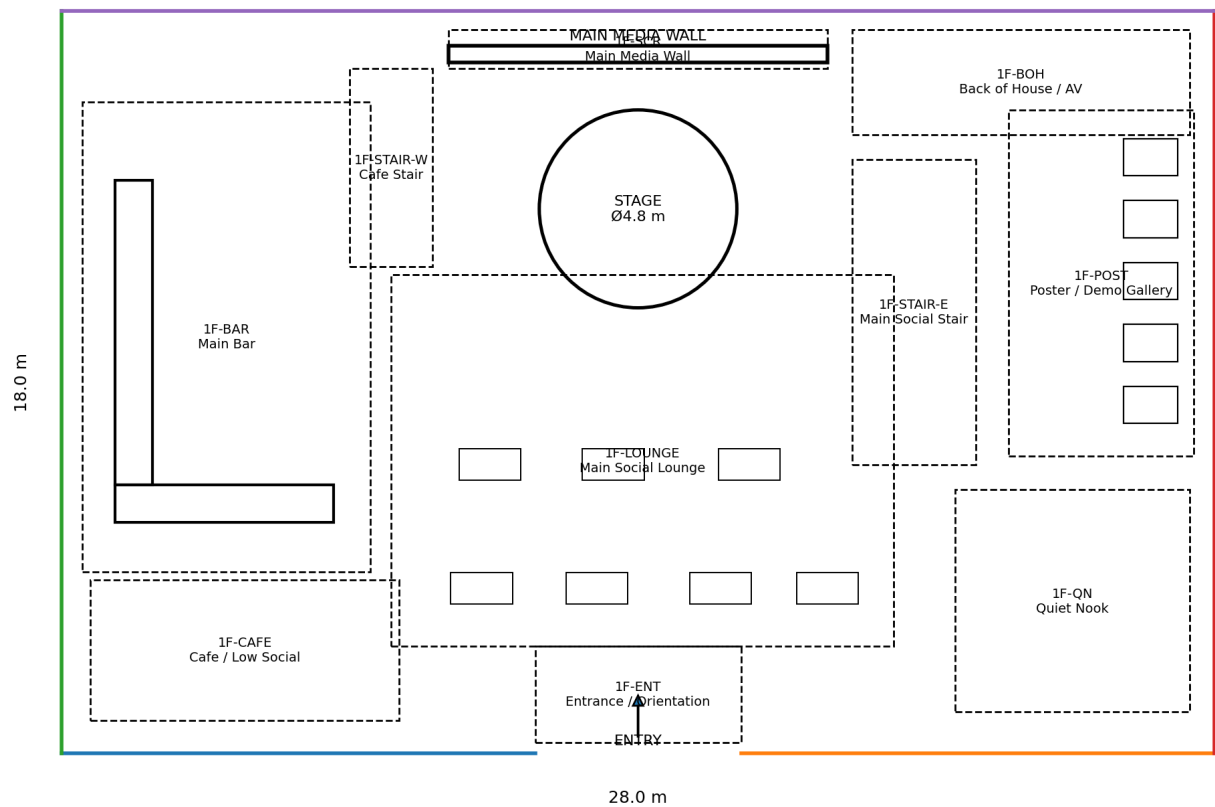
1. Arrival without immediate social pressure
2. Instant visual orientation to stage / bar / 2F / skyline
3. Low-risk entry through cafe, posters or bar
4. Participation in talk, Q&A or central lounge
5. Deeper conversation in mezzanine / terrace / Quiet Room
6. Late-night reduced-light mode

3. General arrangement

CAD units are millimeters. The southwest outside corner is origin (0,0). The 1F datum is +0.000 m. The plans below are concept-development drawings suitable for world blockout, not real-world construction documents.

A-101 | 1F PLAN | THE COMMONS COMPACT | 1:100 concept

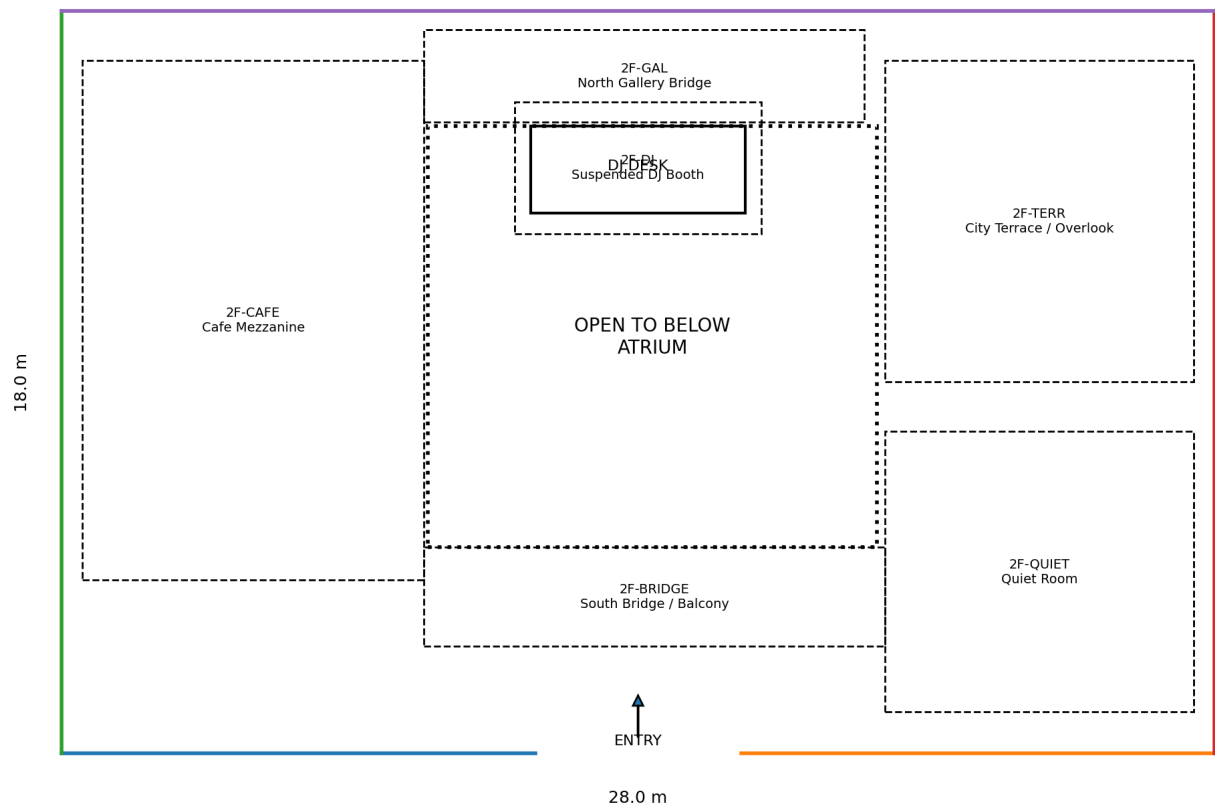
Gross footprint: 28 m x 18 m = 504 m² | 1F +0.000 | Main social / bar / presentation



1F plan

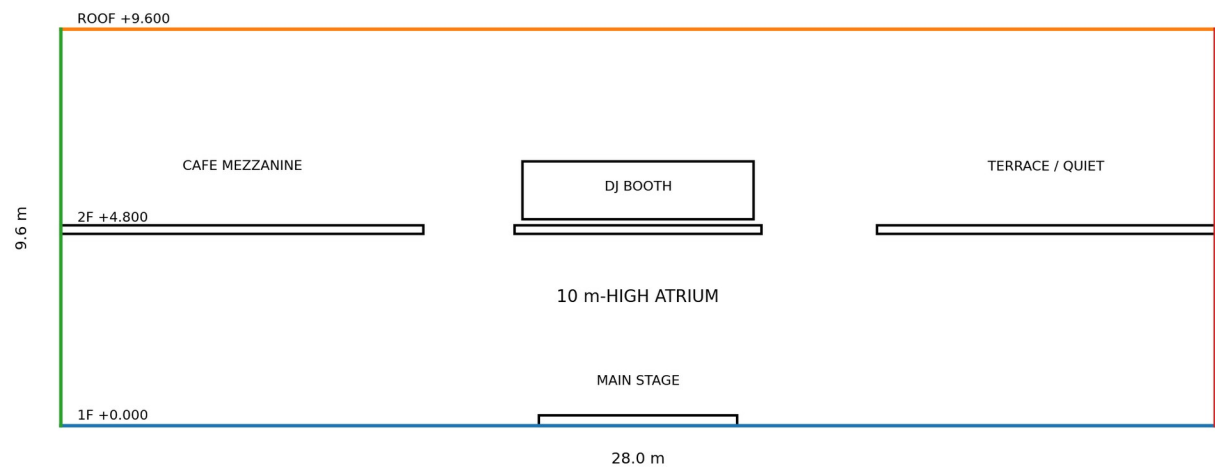
A-102 | 2F MEZZANINE PLAN | THE COMMONS COMPACT | 1:100 concept

2F +4.800 | Cafe mezzanine / DJ / quiet room / terrace | central atrium retained



2F mezzanine plan

A-201 | SECTION A-A | THE COMMONS COMPACT | 1:100 concept



Section A-A

4. Room / zone schedule

ID	Floor	Zone	Approx. area	Capacity
1F-ENT	1F	Entrance / Orientation	11.8 m ²	8
1F-BAR	1F	Main Bar	79.8 m ²	12
1F-CAFE	1F	Cafe / Low Social	25.5 m ²	12
1F-LOUNGE	1F	Main Social Lounge	109.8 m ²	24
1F-STG	1F	Central Stage	18.1 m ²	4
1F-SCR	1F	Main Media Wall	8.7 m ²	0
1F-POST	1F	Poster / Demo Gallery	37.8 m ²	12
1F-QN	1F	Quiet Nook	30.8 m ²	6
1F-STAIR-E	1F	Main Social Stair	22.2 m ²	0
1F-STAIR-W	1F	Cafe Stair	9.6 m ²	0
1F-BOH	1F	Back of House / AV	20.9 m ²	4
2F-CAFE	2F	Cafe Mezzanine	104.6 m ²	18
2F-GAL	2F	North Gallery Bridge	24.1 m ²	10
2F-DJ	2F	Suspended DJ Booth	19.2 m ²	4
2F-TERR	2F	City Terrace / Overlook	58.5 m ²	16
2F-QUIET	2F	Quiet Room	51.0 m ²	8
2F-BRIDGE	2F	South Bridge / Balcony	26.9 m ²	10

5. 1F design intent

- Entrance: 5 m-wide orientation buffer. Spawn users where they can observe before speaking.
- Bar: main everyday social anchor with 10-12 usable stool positions.
- Cafe edge: small tables near entry for low-pressure conversations and waiting.
- Main lounge: modular furniture; clear it for academic seating or DJ floor.
- Stage: Ø4.8 m, 250 mm high. Use stable world-space presenter markers.
- Media wall: 7.1 x 4.0 m effective 16:9 visual surface.
- Poster/demo gallery: six bays minimum, each readable without blocking circulation.
- Quiet Nook: semi-open relief zone only; full Quiet Room is upstairs.
- Two visible stair routes create a loop and prevent the upper level from feeling like a dead-end balcony.

6. 2F design intent

- Cafe mezzanine: the “observer” layer. Users can remain socially present without joining the central cluster.
- DJ booth: visually suspended over the atrium, but dim/inactive during academic mode.
- Terrace: skyline-facing conversation zone and decompression space.
- Quiet Room: 4-8 person enclosed room, low music, warm light, suitable for focused discussion.
- Gallery bridge: standing overlook for event overflow and informal post-talk conversation.

7. Operating modes

Lounge

Warm, stable, low-motion default mode. Stage media is ambient.

Academic

DJ system quiet. Stage/screen prioritized. Poster/demo content enabled. Audience layout becomes explicit.

DJ / Live

Lounge furniture visually retracts or clears; DJ booth, ring lighting and media visualizer activate. Reduced-motion option required.

Quiet Night

Media and cyan emission reduced. Hologram motion slows. Warm local practicals dominate.

8. Academic presentation system

- Main screen: approx. 7.1 x 4.0 m effective display.
- 24 seated audience target; 40 around the main floor during event mode.
- Three presenter/panel marks on stage.
- Session timer and presenter-only control panel.
- Stable world-space pointer/reticle instead of a jittery “laser.”
- 3D demo pedestal adjacent to stage for models/interactive objects.
- Poster bays should be content-swappable without rebuilding architecture.
- After a talk, lighting should automatically return emphasis to the bar/lounge rather than leaving the room feeling like an empty auditorium.

9. Social / comfort design

- Conversation clusters should naturally land at 3-6 people.
- Keep primary pathways 1.8 m clear and secondary routes 1.2 m clear.
- Avoid forced seating and narrow booth traps.
- Use simplified ramp colliders on stairs if required for comfortable locomotion.
- Provide a vertical portal/elevator as an alternate 2F route.
- Comfort controls: bloom/emission, ambient motion, mirror, DJ visual intensity.
- Do not auto-open menus or overlays at spawn.

10. Flat-light shader direction

The architectural style is not photoreal PBR. It uses baked lighting, strong value grouping and a lightly quantized diffuse response so surfaces remain readable and graphic in VR. Detail should come from silhouette, trim, signage, furniture, local emission and baked AO rather than noisy normal maps.

- Opaque master shader: 3 broad diffuse bands with softened thresholds.
- Baked AO at restrained strength.

- Broad low-intensity specular; metal only receives stronger highlights.
- Low-amplitude normal maps.
- Emission is independent of stepped diffuse and clamped before bloom.
- Glass uses a cheap smoked transparent path on PC and a mostly opaque/dither fallback on Quest.
- Hologram uses Fresnel/scanline/slow rotation on PC; single additive layer on Quest.
- Foliage uses cutout material; wind is optional PC-only.

11. Lighting direction

- Bar/cafe practicals: visually 2700-3000 K.
- Stage: neutral 4000-4500 K.
- Cyan/blue reserved for media/music/navigation.
- Architecture should work with baked GI and 0-1 real-time shadowed light in normal mode.
- Use baked reflection probes/cubemaps rather than relying on real-time reflections.
- Never strobe the whole room. DJ changes should remain legible and include reduced-motion behavior.

12. Audio design

- Background music is intentionally under conversation in Lounge mode.
- Do not fake a crowded bar with loud looping chatter.
- Use subtle glass, room tone and city ambience with long non-obvious loops.
- Quiet Room receives strong BGM attenuation.
- DJ energy is reinforced through light and media rather than simply raising SPL.
- Treat bar, lounge, terrace and Quiet Room as distinct conversational clusters.

13. Optimization targets

These are project targets rather than platform hard limits; profile against the current VRChat SDK and actual target devices.

Area	PC target	Quest/mobile target
Visible world geometry	~0.8-1.2M triangles after LOD/occlusion	~200-300k triangles
Real-time shadowed lights	0-1 normal mode	0
Reflections	Baked probes/cubemaps	Cubemap only
Transparency	Limited glass + hologram	Minimize; opaque/dither fallback
Mirrors	Local toggle, default off	Prefer disabled
Texture strategy	2K hero / 1K common / atlases	1K or below / aggressive atlases

14. Work build order

1. Import DXF or rebuild exact 28 x 18 m blockout.
2. Validate scale in VR with several avatar heights.
3. Implement both stairs plus optional vertical portal.

4. Build stage/screen, basic bar, cafe and 2F overlooks.
5. Run 10-20 user social test before detail art.
6. Implement mode controller and academic tools.
7. Create flat-light material library and bake lighting.
8. Add DJ system / AudioLink visuals with comfort controls.
9. Add posters, plants, skyline and hologram.
10. Profile and build Quest fallback.
11. Run 32-48 person event test and adjust audio/social distances.

15. Asset / layer conventions

- Architecture: ARCH_*
- Furniture: FURN_*
- AV / stage: AV_*
- Interactive: INT_*
- Lighting: LGT_*
- Audio: AUD_*
- Optimization/occluder: OPT_*
- Collider-only: COL_*
- Quest fallback meshes/materials: *_Q
- LOD chain: *_LOD0 / *_LOD1 / *_LOD2

End of v0.1 handoff. Geometry and feature changes should update world_spec.json and the CAD drawings first so all downstream art remains synchronized.